Fusion-JEPA: Expressive, Low-Resource Text-to-Speech

via Joint-Embedding Predictive Architecture, MM-DiT, and Continuous Flow Matching

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Motivation: Challenges in Arabic Speech Synthesis

Linguistic Complexity of Arabic

- **Unvocalized Script:** Lack of diacritics (*Tashkeel*) causes heavy pronunciation ambiguity.
- **Rich Morphology:** Complex geminates (*Shaddah*) and emphatic consonants.
- **Prosody:** High sensitivity to pitch and syllable timing.

The Low-Resource Dilemma

- **Data Scarcity:** Studio Arabic speech data is scarce (< 4 hours).
- **Oversmoothing:** Direct L_1/L_2 regression produces muffled, indistinct formant outputs.
- **AR Instability:** Autoregressive models suffer from repetition and attention collapse.

Core Research Question

Can self-supervised latent prediction combined with continuous flow matching achieve stable, high-fidelity speech synthesis without explicit alignment extractors?

Core Solution: The Fusion-JEPA Paradigm

Fundamental Principle: Decoupling Semantics from Acoustics

Rather than forcing the network to memorize noisy, high-frequency pixel intensities, **Fusion-JEPA** splits speech synthesis into two collaborative objectives:

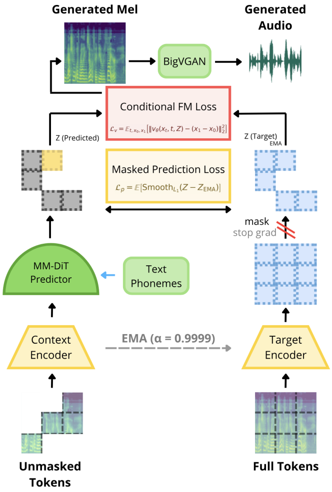
1. Self-Supervised Latent Prediction

- Predicts masked representations inside an abstract, **EMA-updated momentum space**.
- Discards localized acoustic noise and phase variance.
- Learns robust, invariant phonetic features.

2. Continuous Flow Matching

- Denoises Gaussian noise into crisp mel-spectrograms along straight probability paths.
- Reconstructs sharp harmonic boundaries (44.1kHz).
- Organically learns phonetic-acoustic alignments.

Fusion-JEPA: Multimodal Architecture



Key Architectural Components

1. **Context Encoder (ϕ):** Encodes visible acoustic patches alongside phoneme sequence tokens.
2. **Target Encoder ($\bar{\phi}$):** An Exponential Moving Average (EMA) teacher network producing targets $g_i = \mathrm{sg}(\bar{\phi}(U))$.
3. **Feature Predictor (γ):** Regresses masked latent targets via Smooth L_1 loss ($\beta = 2.0$).
4. **Flow Matching Head (v_θ):** Velocity field regressor generating continuous spectrogram trajectories.

Multimodal Joint Attention (MM-DiT) & 1D RoPE

MM-DiT: Bidirectional Joint Attention

- **Mutual Conditioning:** Replaces unidirectional cross-attention bottlenecks.
- **Joint Attention Matrix:** Audio tokens (X) and text tokens (C) share attention with separate MLPs:

$$K_{\text{joint}} = [K_x \parallel K_c], \quad V_{\text{joint}} = [V_x \parallel V_c]$$

- Mutual conditioning guides acoustic structure and refines phoneme durations dynamically.

1D Rotary Position Embeddings (1D RoPE)

- **Length Invariance:** Relative rotational transformations applied directly to query and key vectors:

$$\mathbf{R}_{\Theta,m}\mathbf{q}_i = \begin{pmatrix} \cos m\theta_i & -\sin m\theta_i \\ \sin m\theta_i & \cos m\theta_i \end{pmatrix} \begin{pmatrix} q_{2i} \\ q_{2i+1} \end{pmatrix}$$

- Attention depends purely on relative distance $(m - n)$, handling variable utterance durations cleanly.

Dual Training Objectives & Collapse Prevention

Unified Training Objective

The total loss is the direct sum of self-supervised latent prediction and generative flow matching:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_v + \mathcal{L}_p$$

1. Conditional Flow Matching ($\mathcal{L}_v$)

Linear velocity field from noise x_0 to target x_1:

$$\mathcal{L}_v = \mathbb{E}_{t,x_0,x_1}\left[\|v_\theta(x_t, t, z_i) - (x_1 - x_0)\|_2^2\right]$$

2. JEPA Latent Prediction ($\mathcal{L}_p$)

Smooth L_1 regression toward EMA targets:

$$\mathcal{L}_p = \mathbb{E}\left[\text{Smooth}_{L_1}(u_\theta(z_i) - g_i)\right]$$

Synergistic Collapse Prevention

$\mathcal{L}_v$ prevents representation collapse without contrastive pairs, while $\mathcal{L}_p$ provides strong phonetic guidance.

Speech Corpora & Acoustic Specifications

Corpus	Language	Clips	Duration	Sample Rate
Nawar Halabi ASC	Arabic (MSA)	$1,813$	$\sim$ 3.7 h	44.1 kHz
LJSpeech	English	$13,100$	$\sim$ 24.0 h	44.1 kHz

Linguistic Text Processing

- **Arabic:** Modern Standard Arabic phonetization via Halabi G2P & Buckwalter tokenization.
- **English:** Standard G2P grapheme-to-phoneme conversion (73 token vocabulary).

Acoustic & Vocoder Pipeline

- **Mel Bins:** 128 bands (native BigVGAN alignment).
- **FFT Setup:** 1024 FFT size, 256 hop length (44.1kHz).
- **Neural Vocoder:** Pretrained **BigVGAN v2** (44.1kHz, Snake activations, anti-aliased convs).

Verification Strategy: The Overfitting Benchmark

Why An Overfitting Benchmark?

Before launching large multi-GPU cluster runs, we isolated a fixed subset of studio utterances (5–8 clips) to rigorously verify architectural viability and gradient health.

What the Benchmark Proves

- **Gradient Health:** Monotonic convergence without vanishing or exploding gradients.
- **Collapse Resistance:** Stable EMA teacher latent space without collapse ($\alpha = 0.9999$).
- **Alignment Capacity:** Confirms MM-DiT learns text-to-acoustic alignments from scratch.

Empirical Outcome

- Near-zero training loss on both English and Arabic.
- Synthesis of **audible, intelligible speech**.
- Clear harmonic formant reconstruction matching ground-truth recordings.

Working Audio Results: English Synthesis (LJSpeech)

- **Benchmark Sample:** LJSpeech Index 108 (44.1kHz audio waveform via BigVGAN).
- **Acoustic Analysis:** Accurately captures vowel formants (F_1, F_2, F_3), plosive bursts, and fricative energy.

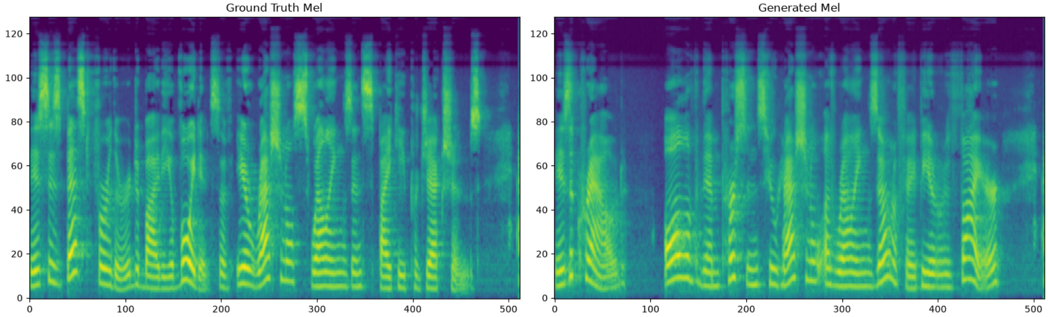


Figure: English Mel-Spectrogram: Ground Truth (Left) vs. Fusion-JEPA Generated Output (Right).

Working Audio Results: Arabic Synthesis (ASC)

- **Benchmark Sample:** Nawar Halabi ASC Index 107 (44.1kHz audio waveform via BigVGAN).
- **Phonetic Fidelity:** Reconstructs Arabic emphatic consonants, gemination (*shaddah*), and natural pause cadence.

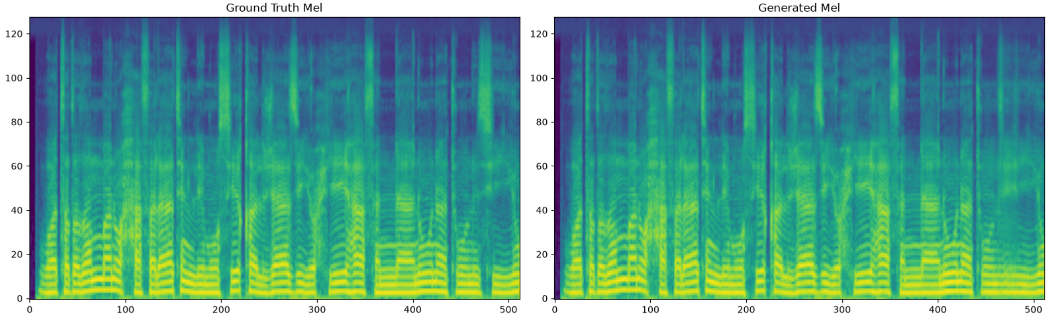


Figure: Arabic Mel-Spectrogram: Ground Truth (Left) vs. Fusion-JEPA Generated Output (Right).

Cross-Platform Infrastructure & Checkpoint Parity

Unified PyTorch Lightning Training Pipeline (`JEPATLightning`)

We engineered a robust, platform-agnostic training infrastructure verified across heterogeneous compute environments:

Google Colab

- NVIDIA T4 GPU (16GB)
- Rapid pipeline debugging and data normalization.

Local Intel XPU

- Intel Arc iGPU (18GB)
- Prototyping with `train_xpu.py`.

KAUST Ibex Cluster

- $4\times$ NVIDIA A100 GPUs
- Slurm submission with Hugging Face syncing.

- **Hardware Header Decoupling:** State dicts cleanly serialized without environment-specific headers (`cuda:0` vs `xpu:0`).

Key Architectural Takeaways

Summary of Technical Innovations

- **First JEPA for Speech Synthesis:** Proved that self-supervised latent prediction natively generalizes to continuous speech synthesis.
- **Elimination of Explicit Aligners:** Non-autoregressive Flow Matching organically learns phonetic-to-acoustic alignments without external duration/pitch predictors.
- **No Spectral Oversmoothing:** Latent space conditioning resolves the regression-to-the-mean failure that plagues standard L_1/L_2 models.
- **Bidirectional Contextualization:** MM-DiT joint attention enables mutual feature refinement between text and acoustic representations.

Project Roadmap & Future Directions

Immediate Next Steps

- **Full Cluster Execution:** Launch full-scale distributed training on KAUST Ibex ($134,000$ steps across $4\times$ A100 GPUs).
- **Formal Objective Benchmarking:** Compute Mel-Cepstral Distortion (MCD) and ASR Word Error Rate (WER) via Whisper.
- **Subjective Evaluation:** Conduct Mean Opinion Score (MOS) listening tests.

Long-Term Research Extensions

- **End-to-End Implicit Diacritization:** Feed raw unvocalized Arabic graphemes directly into MM-DiT to learn joint Tashkeel disambiguation.
- **Multi-Speaker & Dialectal Scaling:** Extend to regional Arabic dialects (Gulf, Egyptian, Levantine) via speaker latent embeddings.

Conclusion & Acknowledgments

Conclusion

Fusion-JEPA demonstrates that coupling Joint-Embedding Predictive Architectures with Multimodal Diffusion Transformers and Flow Matching provides a mathematically principled, highly sample-efficient paradigm for expressive neural text-to-speech in low-resource languages.

Acknowledgments

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Thank You!

Questions & Discussion

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Code Repository: `https://github.com/OmarA32/Fusion-JEPA-TTS`

Interactive Demo: `https://omara32.github.io/Fusion-JEPA-TTS/`